

LAKE MURRAY, SC, USA

WMO: 998177

Lat: 34.1074N

Lon: 81.2707W

Elev: 358

StdP: 14.51

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 25.8	(c) 29.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 25.0	(k) 52.8	(l) 23.2	(m) 55.1	(n) 4.3	(o) 20	(p) 0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.2	(c) 94.5	(d) N/A	(e) 92.5	(f) N/A	(g) 90.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.2	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.5	(b) 16.8	(c) 14.4	(e) 21.5	(f) 99.8	(g) 5.5	(h) 3.2	(i) 17.6	(j) 102.1	(k) 14.3	(l) 104.0	(m) 11.2	(n) 105.8	(o) 7.2	(p) 108.1		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.3	46.0	49.7	55.3	65.2	73.8	80.8	82.6	81.1	76.5	66.0	55.1	50.9		
	DBStd	14.68	9.24	8.82	8.37	6.91	6.15	4.22	3.40	3.98	5.56	7.05	7.39	8.58		
	HDD50	473	190	104	44	2	0	0	0	0	0	1	36	96		
	HDD65	2249	590	431	317	82	12	0	0	0	2	72	302	441		
	CDD50	6070	65	97	207	458	738	924	1011	965	795	498	188	124		
	CDD65	2369	0	4	15	88	284	474	546	500	347	104	4	3		
	CDH74	20970	0	10	57	407	1934	4571	5781	4933	2769	495	11	2		
	CDH80	7866	0	1	4	69	590	1839	2396	1941	925	100	1	0		
(14) Wind	WSAvg	5.3	5.9	5.8	5.8	6.0	5.9	5.9	6.1	4.5	4.1	4.6	4.3	5.2		
(15) Precipitation	PrecAvg	45.6	4.0	3.7	4.4	2.8	2.9	4.8	4.4	4.5	3.8	3.5	3.1	3.5		
	PrecMax	56.7	8.5	7.3	9.6	7.1	5.7	9.4	9.1	8.0	8.9	11.1	7.1	8.9		
	PrecMin	30.5	1.6	1.9	1.1	0.6	0.8	1.4	1.7	1.2	0.2	0.0	1.0	1.2		
	PrecStd	7.4	1.8	1.4	2.0	1.7	1.3	2.4	2.1	2.0	2.2	2.5	1.7	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.2	74.8	80.0	86.3	93.4	97.2	96.5	96.3	93.1	88.3	74.7	72.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	65.5	69.4	74.9	81.6	88.8	93.8	94.4	93.2	90.3	82.4	71.2	69.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	62.4	65.9	71.2	78.6	86.0	91.1	92.1	91.1	88.0	78.9	68.8	66.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	59.6	62.7	67.5	75.6	83.2	88.6	89.7	88.8	85.6	76.3	66.4	64.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	16.6	17.6	17.6	16.4	15.9	15.2	15.4	15.2	16.8	17.4	15.0		
		MCDBR	17.9	19.6	21.8	20.7	19.2	18.7	17.8	18.3	18.3	19.1	20.4	15.7		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.330	0.369	0.410	0.473	0.503	0.550	0.526	0.444	0.399	0.350	0.321		
	taud		2.494	2.467	2.388	2.263	2.136	2.094	1.984	2.039	2.254	2.348	2.444	2.530		
	Ebn at Noon		282	292	287	279	261	252	239	243	259	263	268	276		
	Edh at Noon		27	31	36	43	49	51	57	53	41	34	28	25		
(44) All-Sky Solar Radiation	RadAvg		874	1073	1415	1779	1945	2032	1923	1755	1512	1259	961	758		
	RadStd		46	115	104	133	153	144	128	86	131	151	76	71		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+317	+218

Nomenclature: See separate page